Multiple IV Infusions & Multiple Oral Practice 2

Module 6 - Multiple Dosings · Daily practice sheet · No date

Name _____ Date _____

1a 1. A 300-mg dose of an antibiotic was administered as an IV infusion to a 85 kg male over a period of 90 minutes. Eight hours after the start of the first infusion, a second 300-mg dose was infused, again over a period of 90 minutes. The drug has a half-life of 3.3 hours and apparent volume of distribution of 0.2 L/kg. a. What is the plasma drug concentration at the end of the first infusion?

1b 1. A 300-mg dose of an antibiotic was administered as an IV infusion to a 85 kg male over a period of 90 minutes. Eight hours after the start of the first infusion, a second 300-mg dose was infused, again over a period of 90 minutes. The drug has a half-life of 3.3 hours and apparent volume of distribution of 0.2 L/kg. b. What is the plasma drug concentration 4 hours after the cessation of the second infusion?

2a 2. An adult male (72 kg) was given 300 mg of an antibiotic orally every 8 hours for 2 weeks. The literature reports that the antibiotic is about 85% bioavailable and has a V_D of 3.2 L/kg. The elimination half-life is about 7.3 hours and the absorption half-life is about 75 minutes. Calculate a. C_{max} after the first dose.

2b 2. An adult male (72 kg) was given 300 mg of an antibiotic orally every 8 hours for 2 weeks. The literature reports that the antibiotic is about 85% bioavailable and has a V_D of 3.2 L/kg. The elimination half-life is about 7.3 hours and the absorption half-life is about 75 minutes. Calculate b. C_{min} after the first dose.

2c 2. An adult male (72 kg) was given 300 mg of an antibiotic orally every 8 hours for 2 weeks. The literature reports that the antibiotic is about 85% bioavailable and has a V_D of 3.2 L/kg. The elimination half-life is about 7.3 hours and the absorption half-life is about 75 minutes. Calculate c. Maximum plasma drug concentration at steady state.

2d 2. An adult male (72 kg) was given 300 mg of an antibiotic orally every 8 hours for 2 weeks. The literature reports that the antibiotic is about 85% bioavailable and has a V_D of 3.2 L/kg. The elimination half-life is about 7.3 hours and the absorption half-life is about 75 minutes. Calculate d. Minimum plasma drug concentration at steady state.

2e 2. An adult male (72 kg) was given 300 mg of an antibiotic orally every 8 hours for 2 weeks. The literature reports that the antibiotic is about 85% bioavailable and has a V_D of 3.2 L/kg. The elimination half-life is about 7.3 hours and the absorption half-life is about 75 minutes. Calculate e. Average plasma drug concentration at steady state.